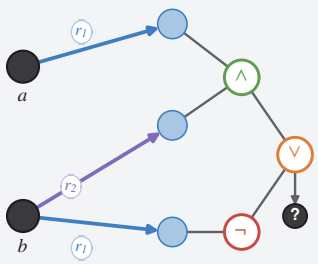
EPFO query

plan over [A, P, I, U, N] (Def. 1)



A P I U N *algebra*

r_1

r_2

negation → complement (N)

Witness evidence

anchored homomorphism counts

relation-interaction graph G_R

branch	witness multiset W_i	
b_1		9
b_2		7
b_3		4

match counts per branch

sufficient statistic (Thm 1)

separates what scores cannot (Thm 2)

Witness-gated algebra

SQA · identifier-free · ~15k params

$$\theta_i = c_0 + c_1\,\sigma(g(s_i^{\max},\ \bar{w}_i,\ \rho_i))$$

$$T_\Theta = \exp[-(\sum_i(-\ln s_i)^{\theta_i})^{1/\theta_t}]$$

$\theta \equiv 1 \Rightarrow$ *exact fixed operators (Lemma 1)*

Thm 2: separation | **Thm 3: symmetric agg.**

Aczél–Alsina family

order-2 evidence: ρ_i from G_R

learnable AND transferable